Linear Algebra midterm 2018 Fall *Name* :

1. Find the solution set of the matrix equation $\mathbf{Ax} = \mathbf{b}$ where

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & -3 & 3 & -1 \\ -2 & -3 & 4 & -7 & 5 \\ 1 & 4 & -7 & 2 & 3 \\ 2 & 5 & -8 & 2 & 7 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 8 \\ 9 \end{bmatrix}.$$

Use the fact that the augmented matrix $\mathbf{M} = [\ \mathbf{A}\ \vdots\ \mathbf{b}\ ]$, its echelon form $\mathbf{U}$, and its reduced echelon form $\mathbf{R}$ are as follows.

$$\mathbf{M} = \left[\begin{array}{rrrrrr} 1 & 2 & -3 & 3 & -1 & 1 \\ -2 & -3 & 4 & -7 & 5 & 2 \\ 1 & 4 & -7 & 2 & 3 & 8 \\ 2 & 5 & -8 & 2 & 7 & 9 \end{array}\right] \xrightarrow{\text{Echelon}} \mathbf{U} = \left[\begin{array}{rrrrrr} 1 & 2 & -3 & 3 & -1 & 1 \\ 0 & 1 & -2 & -1 & 3 & 4 \\ 0 & 0 & 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{\substack{\text{Reduced} \\ \text{Echelon}}} \mathbf{R} = \left[\begin{array}{rrrrrr} 1 & 0 & 1 & 0 & 3 & -2 \\ 0 & 1 & -2 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

The solution can easily be found by solving $\mathbf{R} \cdot [\, x_1\,,\ x_2\,,\ x_3\,,\ x_4\,,\ x_5\,,\ -1\,]^T = \mathbf{0}$ as follows.

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 3 & -2 \\ 0 & 1 & -2 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \qquad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -3x_5 - x_3 - 2 \\ -x_5 + 2x_3 + 3 \\ x_3 \\ 2x_5 - 1 \\ x_5 \end{bmatrix}$$

2. Let the matrix $\mathbf{A}$, its echelon form $\mathbf{U}$, and its reduced echelon form $\mathbf{R}$ be as follows.

$$\mathbf{A} = \left[\begin{array}{rrrrr} 1 & 2 & -3 & 3 & -1 \\ -2 & -3 & 4 & -7 & 5 \\ 1 & 4 & -7 & 2 & 3 \\ 2 & 5 & -8 & 2 & 7 \end{array}\right] \xrightarrow{\text{Echelon}} \mathbf{U} = \left[\begin{array}{rrrrr} 1 & 2 & -3 & 3 & -1 \\ 0 & 1 & -2 & -1 & 3 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{\substack{\text{Reduced} \\ \text{Echelon}}} \mathbf{R} = \left[\begin{array}{rrrrr} 1 & 0 & 1 & 0 & 3 \\ 0 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

a) Solve the matrix equation $\mathbf{Ax} = \mathbf{0}$ where $\mathbf{x} = [x_1\,,\ x_2\,,\ x_3\,,\ x_4\,,\ x_5]^T$.

The solution can easily be found by solving $\mathbf{R} \cdot [\, x_1\,,\ x_2\,,\ x_3\,,\ x_4\,,\ x_5\,]^T = \mathbf{0}$ as follows.

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 3 \\ 0 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \qquad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -3x_5 - x_3 \\ -x_5 + 2x_3 \\ x_3 \\ 2x_5 \\ x_5 \end{bmatrix}$$

b) Find the column space of the matrix $\mathbf{A}$ and its dimension.

$$\text{col}(\mathbf{A}) = \alpha \begin{bmatrix} 1 \\ -2 \\ 1 \\ 2 \end{bmatrix} + \beta \begin{bmatrix} 2 \\ -3 \\ 4 \\ 5 \end{bmatrix} + \gamma \begin{bmatrix} 3 \\ -7 \\ 2 \\ 2 \end{bmatrix}$$

$\dim(\text{col}(\mathbf{A})) = 3$

c) Find the null space of the matrix $\mathbf{A}$ and its dimension.

$$\text{null}(\mathbf{A}) = x_3 \begin{bmatrix} -1 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ -1 \\ 0 \\ 2 \\ 1 \end{bmatrix}$$

$\dim(\text{null}(\mathbf{A})) = 2$

d) Obtain the vectors $\mathbf{u}_i$ from $\mathbf{U}$ such that they span the row space $\text{row}(\mathbf{A})$, i.e. $\text{row}(\mathbf{A}) = \sum_i \alpha_i \mathbf{u}_i$.
Obtain the vectors $\mathbf{r}_i$ from $\mathbf{R}$ such that they span the row space $\text{row}(\mathbf{A})$, i.e. $\text{row}(\mathbf{A}) = \sum_i \beta_i \mathbf{r}_i$.
Find the dimension of the row space $\text{row}(\mathbf{A})$ and the rank of the matrix $\mathbf{A}$.

$$\text{row}(\mathbf{A}) = \alpha_1 \begin{bmatrix} 1 \\ 2 \\ -3 \\ 3 \\ -1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 1 \\ -2 \\ -1 \\ 3 \end{bmatrix} + \alpha_3 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ -2 \end{bmatrix} = \beta_1 \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 3 \end{bmatrix} + \beta_2 \begin{bmatrix} 0 \\ 1 \\ -2 \\ 0 \\ 1 \end{bmatrix} + \beta_3 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ -2 \end{bmatrix}$$

$\dim(\text{row}(\mathbf{A})) = 3$ $\qquad$ $\text{rank}(\mathbf{A}) = 3$

3. Let the matrix $\mathbf{A}$, its echelon form $\mathbf{U}$, and its reduced echelon form $\mathbf{R}$ be as follows.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & -1 \\ -2 & -3 & -7 & 5 \\ 1 & 4 & 2 & 3 \\ 2 & 5 & 2 & 7 \end{bmatrix} \xrightarrow{\text{Echelon}} \mathbf{U} = \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[\text{Echelon}]{\text{Reduced}} \mathbf{R} = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

a) Consider the column vectors of the reduced echelon form $\mathbf{R}$.
- Use the first three columns as basis.
- Find the 4th column as their linear combination.

Note that the first three columns are the standard basis.

$$\alpha_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ -2 \\ 0 \end{bmatrix}$$

$$\alpha_1 = 3 \quad , \quad \alpha_2 = 1 \quad , \quad \alpha_3 = -2$$

b) Consider the column vectors of the matrix $\mathbf{A}$.
- Use the first three columns as basis.
- Find the 4th column as their linear combination.

Hint: Use the fact that $Ax = 0 \Leftrightarrow Ux = 0 \Leftrightarrow Rx = 0$.

$$\beta_1 \begin{bmatrix} 1 \\ -2 \\ 1 \\ 2 \end{bmatrix} + \beta_2 \begin{bmatrix} 2 \\ -3 \\ 4 \\ 5 \end{bmatrix} + \beta_3 \begin{bmatrix} 3 \\ -7 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 5 \\ 3 \\ 7 \end{bmatrix}$$

$$\beta_1 = \alpha_1 = 3 \ , \quad \beta_2 = \alpha_2 = 1 \ , \quad \beta_3 = \alpha_3 = -2$$

4. Find the length $\|\mathbf{u}\|$ of the vector $\mathbf{u} = [\ 3 \ \ -4 \ \ 12\ ]^T$

$$\mathbf{u}^T \cdot \mathbf{u} = [\ 3 \ \ -4 \ \ 12\ ] \cdot [\ 3 \ \ -4 \ \ 12\ ]^T = 3^2 + (-4)^2 + 12^2 = 169 \implies \|\mathbf{u}\| = \sqrt{169} = 13$$

5. Show that the vectors $\mathbf{u}_1 = [\ 3 \ \ 1 \ \ 1\ ]^T$, $\mathbf{u}_2 = [-1 \ \ 2 \ \ 1\ ]^T$, $\mathbf{u}_3 = [-1 \ \ -4 \ \ 7\ ]^T$ are orthogonal.

Note that there are three distinct pairs of these vectors.

$\{\mathbf{u}_1 \ , \ \mathbf{u}_2\}$
$\{\mathbf{u}_1 \ , \ \mathbf{u}_3\}$
$\{\mathbf{u}_2 \ , \ \mathbf{u}_3\}$

The inner products of these pairs are all zero. Therefore, $\mathbf{u}_1$, $\mathbf{u}_2$, and $\mathbf{u}_3$ are orthogonal.

$$\mathbf{u}_1 \cdot \mathbf{u}_2 = [\ 3 \ \ 1 \ \ 1\ ] \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} = 3 \cdot (-1) + 1 \cdot 2 + 1 \cdot 1 = -3 + 2 + 1 = 0$$

$$\mathbf{u}_1 \cdot \mathbf{u}_3 = [\ 3 \ \ 1 \ \ 1\ ] \begin{bmatrix} -1 \\ -4 \\ 7 \end{bmatrix} = 3 \cdot (-1) + 1 \cdot (-4) + 1 \cdot 7 = -3 - 4 + 7 = 0$$

$$\mathbf{u}_2 \cdot \mathbf{u}_3 = [-1 \ \ 2 \ \ 1\ ] \begin{bmatrix} -1 \\ -4 \\ 7 \end{bmatrix} = -1 \cdot (-1) + 2 \cdot (-4) + 1 \cdot 7 = +1 - 8 + 7 = 0$$

6. Assume that the vectors $\mathbf{u}_1 = [\ 3 \ \ 1 \ \ 1\ ]^T$, $\mathbf{u}_2 = [-1 \ \ 2 \ \ 1\ ]^T$, $\mathbf{u}_3 = [-1 \ \ -4 \ \ 7\ ]^T$ are orthogonal.
Express the vector $\mathbf{y} = [\ 6 \ \ 1 \ \ -8\ ]^T$ as their linear combination, i.e. $\mathbf{y} = \alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \alpha_3\mathbf{u}_3$.
Hint: Because the vectors $\mathbf{u}_1$, $\mathbf{u}_2$, $\mathbf{u}_3$ are orthogonal, the coefficients $\alpha_i = \mathbf{y} \cdot \mathbf{u}_i / (\mathbf{u}_i \cdot \mathbf{u}_i)$.

$$\mathbf{y} \cdot \mathbf{u}_1 = [\ 6 \ \ 1 \ \ -8\ ] \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} = 6 \cdot 3 + 1 \cdot 1 - 8 \cdot 1 = 11$$

$$\mathbf{u}_1 \cdot \mathbf{u}_1 = [\ 3 \ \ 1 \ \ 1\ ] \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} = 3^2 + 1^2 + 1^2 = 11$$

$$\alpha_1 = \tfrac{11}{11} = 1$$

$$\mathbf{y} \cdot \mathbf{u}_2 = [\ 6 \ \ 1 \ \ -8\ ] \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} = 6 \cdot (-1) + 1 \cdot 2 - 8 \cdot 1 = -12$$

$$\mathbf{u}_2 \cdot \mathbf{u}_2 = [\ -1 \ \ 2 \ \ 1\ ] \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} = (-1)^2 + 2^2 + 1^2 = 6$$

$$\alpha_2 = \tfrac{-12}{6} = -2$$

$$\mathbf{y} \cdot \mathbf{u}_3 = [\ 6 \ \ 1 \ \ -8\ ] \begin{bmatrix} -1 \\ -4 \\ 7 \end{bmatrix} = 6 \cdot (-1) + 1 \cdot (-4) - 8 \cdot 7 = -66$$

$$\mathbf{u}_3 \cdot \mathbf{u}_3 = [\ -1 \ \ -4 \ \ 7\ ] \begin{bmatrix} -1 \\ -4 \\ 7 \end{bmatrix} = (-1)^2 + (-4)^2 + 7^2 = 66$$

$$\alpha_3 = \tfrac{-66}{66} = -1$$

$$\mathbf{y} = \alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \alpha_3\mathbf{u}_3 = \mathbf{u}_1 - 2\mathbf{u}_2 - \mathbf{u}_3.$$